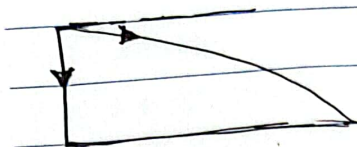


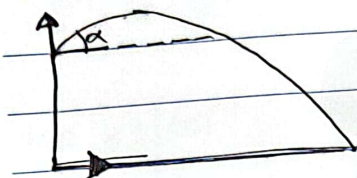
Note



$$x = ut$$

$$y = \frac{1}{2}gt^2$$

$$V_y = u \sin \alpha - gt \quad V_x = u \cos \alpha$$



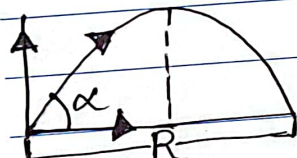
$$x = ut \cos \alpha$$

$$y = -ut \sin \alpha + \frac{1}{2}gt^2$$



$$x = ut \cos \alpha$$

$$y = ut \sin \alpha + \frac{1}{2}gt^2$$

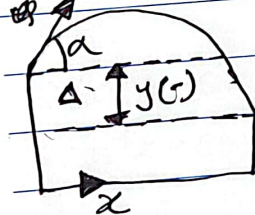


$$x = ut \cos \alpha$$

$$y = ut \sin \alpha - \frac{1}{2}gt^2$$

$$H = \frac{u^2 \sin^2 \alpha}{2g}$$

$$\tan \alpha = \frac{4H}{R}$$



$$t_{\max} = \frac{u \sin \alpha}{g}$$

$$R = \frac{u^2 \sin 2\alpha}{g}$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha}$$

1. Complete the following sentences by writing the correct word or phrase in the space provided.
In an educational institution that is (a) ... (b) ... (c) ... (d) ... (e) ... (f) ... (g) ... (h) ... (i) ... (j) ...
2. In conformity with the circumstances of the case, the court has decided to grant the applicant the sum of ...
3. The court has decided to grant the applicant the sum of ...

Note HM-2 CHPT-7

$$1. \sin^{-1} x + \cos^{-1} x = \pi/2$$

$$\tan^{-1} x + \cot^{-1} x = \pi/2$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \pi/2$$

$$2. \tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$$

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \frac{x+y+z-xyz}{1-xy-yz-zx}$$

$$3. \begin{cases} \sin^{-1} x + \sin^{-1} y = \sin^{-1} \{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} \\ \sin^{-1} x - \sin^{-1} y = \sin^{-1} \{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \end{cases}$$

$$\begin{cases} \cos^{-1} x + \cos^{-1} y = \cos^{-1} \{xy \ominus \sqrt{(1-x^2)(1-y^2)}\} \\ \cos^{-1} x \ominus \cos^{-1} y = \cos^{-1} \{xy \oplus \sqrt{(1-x^2)(1-y^2)}\} \end{cases}$$

$$\begin{cases} \cos^{-1} x \oplus \cos^{-1} y = \cos^{-1} \{xy \ominus \sqrt{(1-x^2)(1-y^2)}\} \\ \cos^{-1} x \ominus \cos^{-1} y = \cos^{-1} \{xy \oplus \sqrt{(1-x^2)(1-y^2)}\} \end{cases}$$

$$4. 2\sin^{-1} x = \sin^{-1} \{2x\sqrt{1-x^2}\}$$

$$2\cos^{-1} x = \cos^{-1} (2x^2 - 1)$$

$$2\tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2}$$

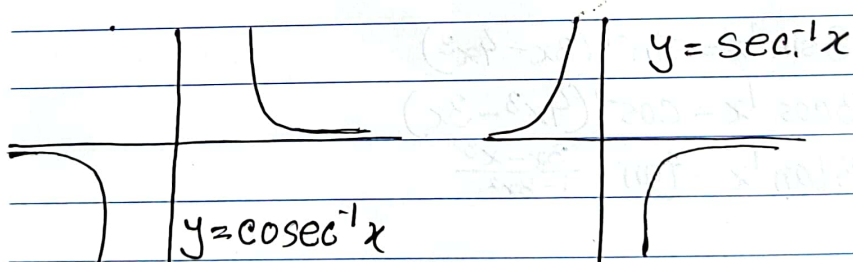
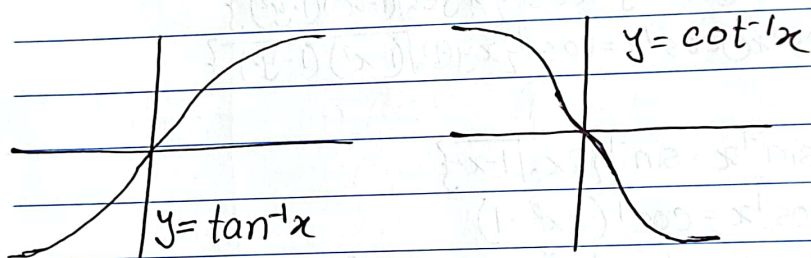
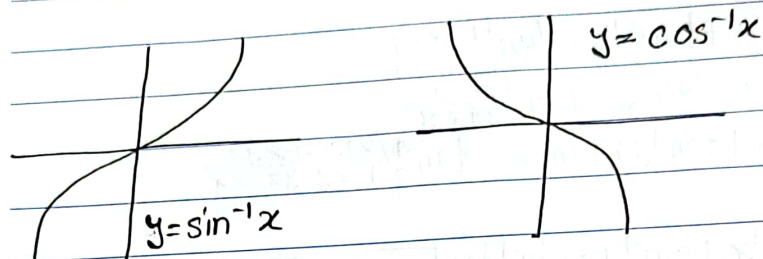
$$5. 3\sin^{-1} x = \sin^{-1} (3x - 4x^3)$$

$$3\cos^{-1} x = \cos^{-1} (4x^3 - 3x)$$

$$3\tan^{-1} x = \tan^{-1} \frac{3x - x^3}{1 - 3x^2}$$

Note

$$\begin{aligned} 6. \sin \theta &= \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha \\ \cos \theta &= \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha \\ \tan \theta &= \tan \alpha \Rightarrow \theta = n\pi + \alpha \end{aligned}$$



Note HM-1

$$1. \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$$

$$2. \sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$$
$$= \cos^2 B - \cos^2 A$$

$$\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B$$
$$= \cos^2 B - \sin^2 A$$

$$3. \sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$
$$= \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Note

$$4. \begin{aligned} 2\sin^2 A &= 1 - \cos 2A \\ 2\cos^2 A &= 1 + \cos 2A \\ \tan^2 A &= \frac{1 - \cos 2A}{1 + \cos 2A} \end{aligned}$$

$$5. \begin{aligned} \sin 3A &= 3\sin A - 4\sin^3 A \\ \cos 3A &= 4\cos^3 A - 3\cos A \\ \tan 3A &= \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A} \end{aligned}$$

$$6. \begin{aligned} 4\sin^3 A &= 3\sin A - \sin 3A \\ 4\cos^3 A &= 3\cos A + \cos 3A \end{aligned}$$

$$7. \begin{aligned} 2\cos A \sin B &= \sin(A+B) - \sin(A-B) \\ 2\sin A \cos B &= \sin(A+B) + \sin(A-B) \\ 2\cos A \cos B &= \cos(A+B) + \cos(A-B) \end{aligned}$$

$$* 2\sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$8. \sin C + \sin D = 2\sin \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\sin C - \sin D = 2\cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

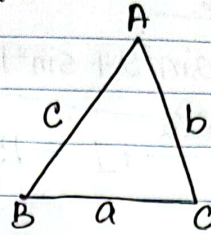
$$\cos C + \cos D = 2\cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\cos C - \cos D = 2\sin \frac{C+D}{2} \sin \frac{D-C}{2}$$

Note

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

ଅନୁପାତ ସମୀକରଣ



$$\cos 2A = \frac{b^2 + c^2 - 2a^2}{2bc}$$

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\Delta = \frac{\sqrt{3}}{4} a^2 \quad \Delta = \frac{b}{4} \sqrt{4a^2 - b^2}$$

$$a = b \cos C + c \cos B$$

ଅନୁପାତ ସମୀକରଣ

$$\Delta = \frac{1}{2} bc \sin A = \frac{abc}{4R}$$

$$\cos 2B = \frac{c^2 + a^2 - 2b^2}{2ca} \quad \cos 2C = \frac{a^2 + b^2 - 2c^2}{2ab}$$

$$\tan \frac{C-A}{2} = \frac{c-a}{c+a} \cot \frac{B}{2} \quad \tan \frac{B-A}{2} = \frac{a-b}{b+a} \tan \frac{C}{2}$$

$$a = a \cos C + c \cos A \quad / \quad c = a \cos B + b \cos A$$

$$\Delta = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ca}$$

1. Complete the following sentences.
Gender studies help us to understand the relationship between gender and social norms.
In an educational institution, teachers should help students to develop their critical thinking skills.
but (e) — help from them (b) — on the (a) — their men (c) — their men (d) — their men

Note

$$\sin^2 3 + \sin^2 9 + \sin^2 15 + \dots + \sin^2 177 = \frac{n}{2}$$

$$n = \frac{177-3}{6} + 1 = 15$$

$$n_{\text{term}} = a + (n-1)d$$

$$\cancel{2n\pi = 0} \text{ হলে, } 2n\theta = \pi \text{ হলে,}$$

$$\begin{aligned} * \tan \theta \cdot \tan 2\theta \cdot \tan 3\theta \cdot \dots \cdot \tan (n-1)\theta \cdot \tan (n+1)\theta \cdot \dots \cdot \tan (2n-1)\theta \\ = (-1)^{n-1} \end{aligned}$$

$$\begin{aligned} * \cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \dots \cdot \cot (n-1)\theta \cdot \cot (n+1)\theta \cdot \dots \cdot \cot (2n-1)\theta \\ = \frac{1}{(-1)^{n-1}} \end{aligned}$$

$$\begin{aligned} * \tan \theta \tan 3\theta \tan 5\theta \cdot \dots \cdot \tan (2n-1)\theta = (-1)^{n/2} \\ \cot () \longrightarrow \frac{1}{(-1)^{n/2}} \end{aligned}$$

$$* \tan \theta \tan 3\theta \tan 5\theta \cdot \dots \cdot \tan (n-1)\theta = \cot(\theta) \rightarrow = 1$$

$$4n\theta = \pi \text{ হলে,}$$

$$\tan \theta \tan 2\theta \tan 3\theta \cdot \dots \cdot \tan (2n-1)\theta = \cot(\theta) \rightarrow = 1$$

Note HM-2 CHPT-3

$$\omega = \frac{-1 + \sqrt{-3}}{2} \quad \omega^2 = \frac{-1 - \sqrt{-3}}{2}$$

$$z = x + iy \quad \bar{z} = x - iy$$

$$|z| = |x + iy| = \sqrt{x^2 + y^2} = |\bar{z}|$$

$$\theta = \tan^{-1} \frac{y}{x} \quad [\theta \leq 180]$$

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

$$\omega^n + \omega^{n+1} + \omega^{n+2} = 0$$

$$\sqrt{n+n\sqrt{n+n\sqrt{\dots}}} = \frac{n \pm \sqrt{n^2 - 4n}}{2}$$

$$\sqrt[4]{-n^2} = \pm \sqrt{n/2} (1 \pm i)$$

$$\sqrt[6]{-n^3} = \pm \sqrt{n} i, \pm \sqrt{n} \omega, \pm \sqrt{n} \omega^2$$

$$Z = r e^{i\theta} = r (\cos \theta + i \sin \theta)$$

$$e^{i\pi} + 1 = 0$$

Note

$$|z-k|=r \quad (O) \quad C=(k,0) \quad \text{rad}=r$$

$$|az+b|=|az+c| \quad (—)$$

$$|z+a| \neq |\bar{z}+b| \quad (—)$$

$$|az+c|=|bz+d| \quad (O)$$

$$z\bar{z}=r \quad (O) \quad C=(0,0) \quad \text{rad}=r$$

$$|z+a| = x \quad / \quad |z+a| = y \quad (+ \in)$$

$$|z+a| + |z-a| = b \quad (O) \quad F=(\pm a,0), \text{ જુડેલક} = b$$

$$|z+a| - |z-a| = b \quad (H) \quad F=(\pm a,0) \text{ જાડે } = b$$

$$\text{પ્રમાણ} \Rightarrow x+iy \Rightarrow r(\cos\theta + i\sin\theta)$$

$$r = \sqrt{x^2+y^2} \quad \theta = \tan^{-1} \frac{y}{x} \quad [\text{Argument}]$$

1. Complete the following sentences
Gender studies are important in understanding the role of women in society.
In conformity with the circumstances, the government has taken steps to improve the living conditions of the poor.
In an educational institution, the teacher's relationship with the students is very important.
but (e) — help from them in — on the teacher's —
they (h) — — — — —
3. they (h) — — — — —
In an educational institution, the teacher's relationship with the students is very important.
but (e) — help from them in — on the teacher's —
they (h) — — — — —

Note

$$i + i^2 + i^3 + \dots i^{2023} \Rightarrow 2023 \% 4 = 3$$

$$\Rightarrow i + i^2 + i^3 = 1 + (-1) - 1 = -1$$

$$\sqrt{a \pm ib} = \pm \frac{1}{\sqrt{2}} (\sqrt{r+a} \pm i \sqrt{r-a}) [r = \sqrt{a^2 + b^2}]$$

$$z = re^{i\theta}$$

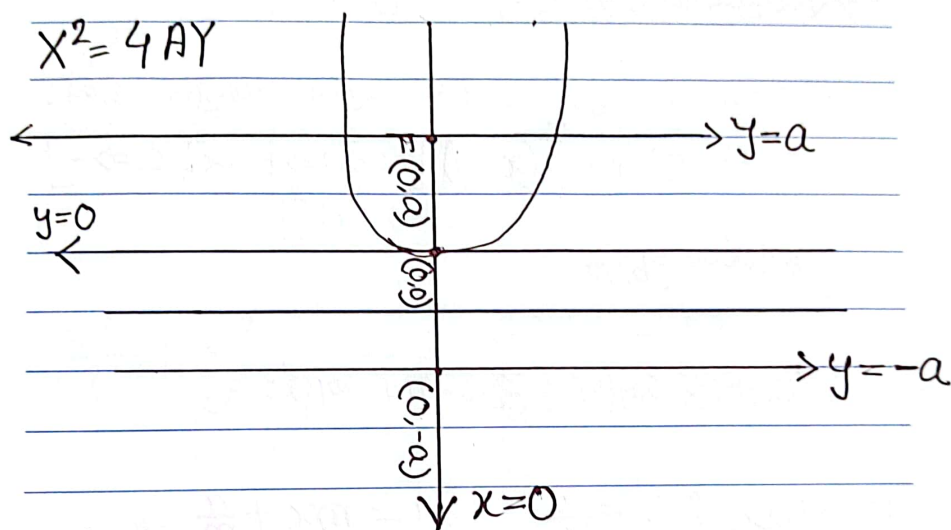
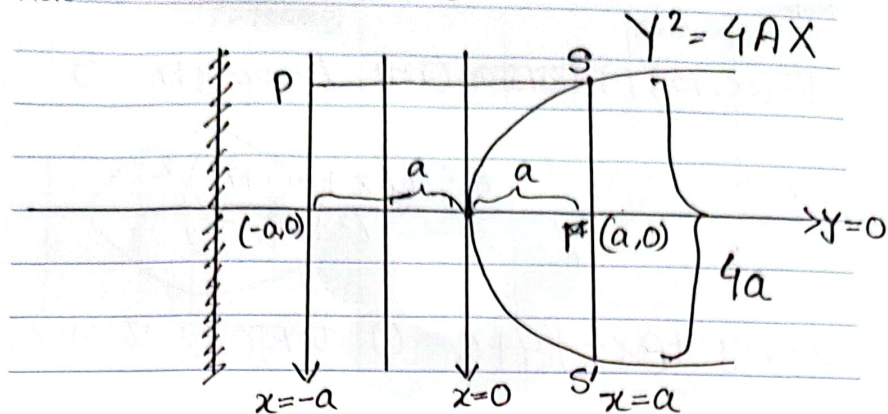
$$\frac{a+ib}{x+iy} = \frac{ax+by}{x^2+y^2} + i \frac{bx-ay}{x^2+y^2}$$

$$a^3 + b^3 + c^3 = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) + 3abc$$

$$\sum a^2b = (a+b+c)(ab+bc+ca) - 3abc$$

Note HM-1

CHPT-6



কোন (α, β) অঙ্কে, $||X||$

$$(Y - \beta)^2 = 4A(X - \alpha)$$

অঙ্কে $Y||$

$$(X - \alpha)^2 = 4A(Y - \beta)$$

1. Complete the following sentences:
Gender studies are the study of the role of gender in society.
In an educational institution that is (b) but (c) relationship that is (b) but (c) help from them in the process of their learning.

Note

$F(\alpha, \beta)$, নিম্নলিখিত রেখা $lx+my+n=0$

$$(x-\alpha)^2 + (y-\beta)^2 = e^2 \left(\frac{lx+my+n}{l^2+m^2} \right)^2$$

$ax^2+by^2+gx+fy+c=0$ উপবৃত্তের উ:ন: দে

$$2\sqrt{a} [a > b] \quad // \quad 2a/\sqrt{b} [b > a]$$

(α, β) কীর্ষ ও (a, b) উপকেন্দ্র হলে পরাবৃত্তের সমী:

$$(y-\beta)^2 = (\pm) |x-a| (x-\alpha) \quad [\alpha < a \Rightarrow + \quad \alpha > a \Rightarrow -]$$

(α, β) কীর্ষ ও (b, β)

$y=mx+c$ স্পর্শক হওয়ার ক্রম:

$$y^2 = 4ax : c = \frac{a}{m} \quad y = mx + \frac{a}{m}$$

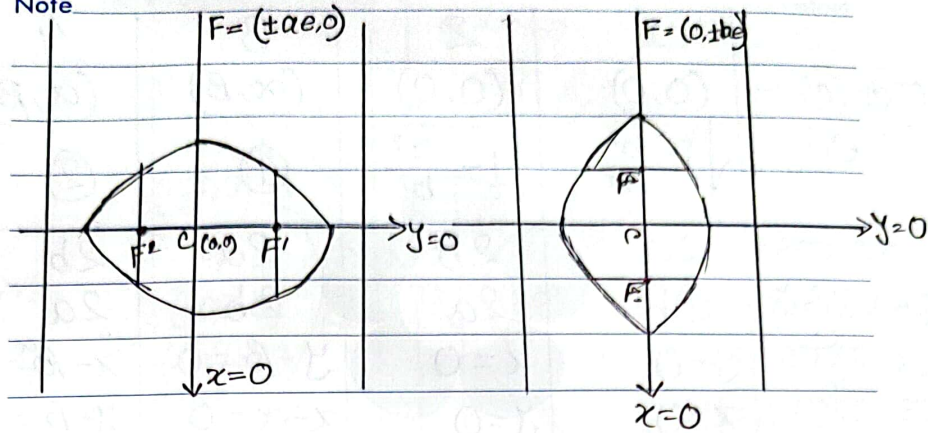
$$x^2 = 4ay : c = -am^2$$

$$a > b$$

$$\text{Area} = \pi ab$$

$$b > a$$

Note



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$ae = \sqrt{a^2 - b^2}$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} < 1$$

$$[b > a]$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} < 1$$

$$[a > b]$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow$$

$$(a > b) - \text{①}$$

$$(b > a) - \text{②}$$

$$\frac{(x-a)^2}{a^2} + \frac{(y-b)^2}{b^2} = 1$$

$$(a > b) - \text{③}$$

$$(a < b) - \text{④}$$

$$P(x, y) \text{ parametric form} = (a \cos \theta, b \sin \theta)$$

$$\text{eccentric angle, } \theta = \tan^{-1} \left(\frac{ay}{bx} \right)$$

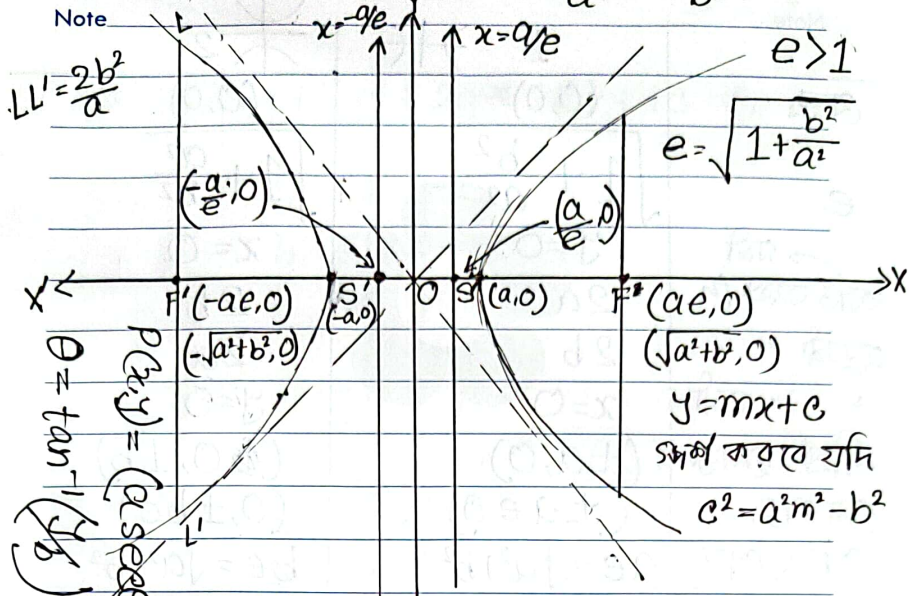
Note	1	2	3	4
କେନ୍ଦ୍ର (c)	(0,0)	(0,0)	(α, β)	(α, β)
e	$\sqrt{1-\frac{b^2}{a^2}}$	$\sqrt{1-\frac{a^2}{b^2}}$	①	②
ବୃହତ୍ ଅକ୍ଷ ଲଂ:	$2a$	$2b$	$2a$	$2b$
ଛୋଟ ଅକ୍ଷ ଲଂ:	$2b$	$2a$	$2b$	$2a$
ବୃହତ୍ ସମୀ:	$y=0$	$x=0$	$y-\beta=0$	$x-\alpha=0$
ଛୋଟ ସମୀ:	$x=0$	$y=0$	$x-\alpha=0$	$y-\beta=0$
କୀର୍ଣ୍ଣ	($\pm a, 0$)	($0, \pm b$)	($\pm a+\alpha, \beta$)	($\alpha, \pm b+\beta$)
ଫୋକାଲ, F	($\pm ae, 0$)	($0, \pm be$)	($\pm ae+\alpha, \beta$)	($\alpha, \pm be+\beta$)
	$ae=\sqrt{a^2-b^2}$	$be=\sqrt{b^2-a^2}$	①	②
ନିର୍ଦ୍ଧାରିତ ପାଦାଂଶୁ	($\pm \frac{a}{e}, 0$)	($0, \pm \frac{b}{e}$)	($\pm \frac{a}{e}+\alpha, \beta$)	($\alpha, \pm \frac{b}{e}+\beta$)
କେନ୍ଦ୍ର ଉପରେ	$\frac{a}{e} =$	$\frac{b}{e} =$	①	②
ନିର୍ଦ୍ଧାରିତ	a^2	b^2		
ପାଦାଂଶୁ ଦୂରତା	$\sqrt{a^2-b^2}$	$\sqrt{b^2-a^2}$		
ନିର୍ଦ୍ଧାରିତ ସମୀ:	$x=\pm a/e$	$y=\pm b/e$	$x-\alpha=\pm a/e$	$y-\beta=\pm b/e$
ଓ: ଲଂ: ଦି:	$2b^2/a$	$2a^2/b$	$2b^2/a$	$2a^2/b$
ଓ: ଲଂ: ସମୀ:	$x=\pm ae$	$y=\pm be$	$x-\alpha=\pm ae$	$y-\beta=\pm be$

~~Prove that~~

① $\rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Note

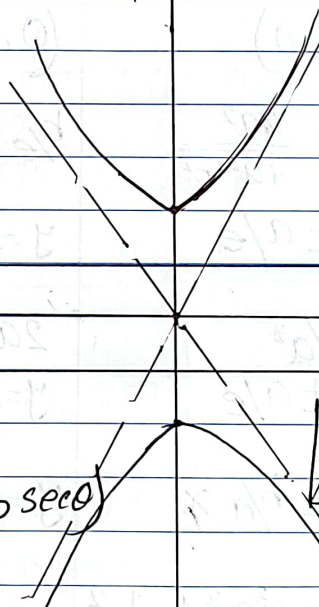
$LL' = \frac{2b^2}{a}$



$P(x, y) = (a \sec \theta, b \tan \theta)$

$\theta = \tan^{-1}(y/x)$

② $\rightarrow \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$



$P(x, y) = (a \tan \theta, b \sec \theta)$

$\theta = x/a$

$\theta = \tan^{-1} x/a$

1. Complete the following sentences:
 In an educational institution, the relationship between the teacher and the student is (a) _____ but (b) _____
 In an educational institution, the relationship between the teacher and the student is (a) _____ but (b) _____
 In an educational institution, the relationship between the teacher and the student is (a) _____ but (b) _____

Note	1 $\rightarrow$ H	2 $\rightarrow$ V
କେନ୍ଦ୍ର	(0,0)	(0,0)
e	$\sqrt{1 + \frac{b^2}{a^2}}$	$\sqrt{1 + \frac{a^2}{b^2}}$
$\rightarrow$ ଯକ୍ଷୀ	$y=0$	$x=0$
ଅର୍ଦ୍ଧ ଯକ୍ଷୀ ଦି	$2a$	$2b$
ଅର୍ଦ୍ଧ ଯକ୍ଷୀ	$2b$	$2a$
$\rightarrow$ ଯକ୍ଷୀ	$x=0$	$y=0$
କ୍ଷୀର ଯକ୍ଷୀ	$(\pm a, 0)$	$(0, \pm b)$
ଯୋଗ୍ୟ	$(\pm ae, 0)$	$(0, \pm be)$
OF & OF'	$ae = \sqrt{a^2 + b^2}$	$be = \sqrt{a^2 + b^2}$
ବିହୀନ ଯକ୍ଷୀ	$(\pm \frac{a}{e}, 0)$	$(0, \pm \frac{b}{e})$
	$a/e = \frac{a^2}{\sqrt{a^2 + b^2}}$	$b/e = \frac{b^2}{\sqrt{a^2 + b^2}}$
n ଯକ୍ଷୀ:	$x = \pm a/e$	$y = \pm b/e$
ଓ: ଯକ୍ଷୀ ଦି:	$2b^2/a$	$2a^2/b$
$\rightarrow$ ଯକ୍ଷୀ:	$x = \pm a/e$	$y = \pm b/e$
ଆସିକାଟ	$y = \pm b/a x$	$y = \pm b/a x$
$y = mx + c$	$c^2 = a^2 m^2 - b^2$	

Note

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$\Delta = 0 \quad (\text{---})$$

$$\Delta \neq 0 \quad a=b \quad h=0 \quad (0)$$

$$\Delta \neq 0 \quad ab-h^2 > 0 \quad (O)$$

$$\Delta \neq 0 \quad ab-h^2 < 0 \quad (\text{---})$$

$$h^2 = ab \quad (\text{---})$$

$$(i) \quad y^2 = 4ax \quad \text{---} \quad y = mx + c \quad [c = a/m]$$

$$\left(\frac{a}{m^2}, \frac{2a}{m} \right)$$

$$(ii) \quad x^2 = 4ay \quad \text{---} \quad y = mx + c \quad [c = -am^2]$$

$$(2am, am^2)$$

$$y^2 = ax + by + c \quad \text{at } (x, y) \quad y = b/2$$

$$x^2 = ax + by + c \Rightarrow x = a/2 \quad \downarrow$$

$$\downarrow y = ? \quad x = ?$$

Note HM-1

Chpt-4

① $x^2 + y^2 + 2gx + 2fy + c = 0$

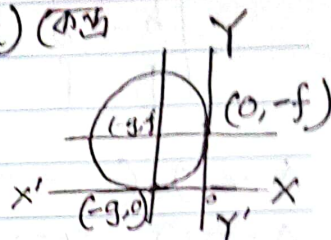
$\bullet C = (-g, -f)$

$r = \sqrt{g^2 + f^2 - c}$

$(x-h)^2 + (y-k)^2 = r^2$ (h, k) (কেন্দ্র)

② x অক্ষের উপর $2\sqrt{g^2 - c}$

y অক্ষের উপর $2\sqrt{f^2 - c}$



③ $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

$(x_1, y_1), (x_2, y_2)$ চারদিক

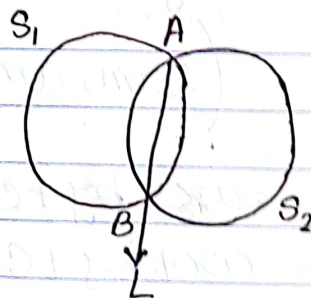
④ ক্ষেত্রের ক্ষেত্রফল $= \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$

⑤ বহিঃস্থ বিন্দু থেকে ক্ষেত্রফল

$(x^2 + y^2 + 2gx + 2fy + c)(x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c)$

$= \{xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c\}^2$

$SS_1 = T^2$



6. A, B বিন্দু গামী চোলা

বৃত্তের সমীকরণ,

$S_1 \pm KL$

(12)

(x_1, y_1)

ପିନ୍ଧୁତେ

ਸ਼੍ਰੀਮਤੀ ਜਸਮੀਤ ਕੌਰ
ਸ਼੍ਰੀਮਤੀ ਜਸਮੀਤ ਕੌਰ

$$\left(\frac{1}{2} \right)^n$$

$(x, y,)$

ആദ്യ ഖണ്ഡം

Chpt-3
 $f(x) = ax + by + c = 0$ এর সাথে $g(x) = a_1x + b_1y + c_1 = 0$
 এর প্রতিবন্ধ্য:
 $(a^2 + b^2)g(x) - 2(aa_1 + bb_1)f(x) = 0$

$$f(x) = ax + by + c = 0 \text{ and } g(x) = a_1x + b_1y + c_1 = 0$$

এর প্রতিবিম্ব:

$$(a^2 + b^2)g(x) - 2(aa_1 + bb_1)f(x) = 0$$

★ **প্রাক্ষিপিক:** $A(x_1, y_1)$ ও $B(x_2, y_2)$ বিন্দুগামী হ্রদ সমী:

$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) +$$

$$K \{ (x-x_1)(y-y_2) - (y-y_1)(x_1-x_2) \} = 0$$

~~C~~ $C(x, y)$ হলে হকের (দানার সমী, ব্যাসের R হলে

$$R^2 = r^2 + r_1^2 - 2rr_1 \cos(\theta - \theta_1)$$

1. Complete the sentences.

Gender roles are different in different cultures.

2. Put the words in the correct order to make sentences.

our total set-up (a) is conformity with the circumstances of the country

relationships that is (b) help from them on the teacher-student relationship

but (c) they are different in different cultures

they (d) are different in different cultures

understand (e) an educational institution form consider the circumstances of the country

Note HM-1

Chpt 3

$$① (x, y) \Rightarrow (r, \theta)$$

$$x = r \cos \theta, y = r \sin \theta \quad \theta = \tan^{-1}(y/x)$$

$$r = \sqrt{x^2 + y^2}$$

$$② P(x_1, y_1) \quad Q(x_2, y_2)$$

$$PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$m_1 : m_2$ অৱপাত অকুৰিডকালী বিন্দু

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_2 y_1 - m_1 y_2}{m_1 - m_2} \right) \text{ বহিৰ্বিডক}$$

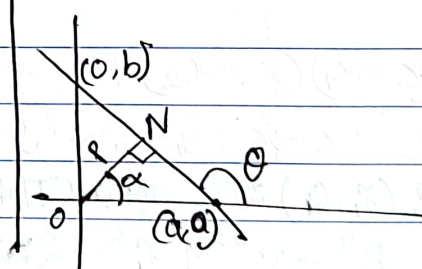
$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$x/a + y/b = 1$$

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta}$$

$$x \cos \alpha + y \sin \alpha = p$$



Note

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}$$

দুটি রেখার মধ্যবর্তী কোণ
(+) ধনাত্মক কোণ (-) ঋণাত্মক কোণ

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad (x_1, y_1) \text{ বিন্দু ও } ax + by + c = 0 \text{ এর দূরত্ব}$$

$$\frac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}} \quad \left| \begin{array}{l} \text{সমান্বিত্বক্ষেপের} \\ \text{সমীকরণ} \end{array} \right.$$

* c_1 ও c_2 উভয় (+) / (-) হলে (+) চিহ্নের সমীকরণ
* মূল বিন্দু দিতে।

* $a_1 a_2 + b_1 b_2 > 0$ হলে (+) চিহ্নের সমীকরণ ~~সুস্থ~~ মূল
* কোণের সমান্বিত্বক্ষেপ।

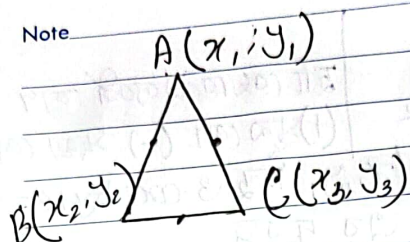
$$\left. \begin{array}{l} a_1 x + b_1 y + c_1 = 0 \\ a_2 x + b_2 y + c_2 = 0 \end{array} \right\} \begin{array}{l} x \\ y \end{array} = \frac{b_1 c_2 - b_2 c_1}{c_1 a_2 - c_2 a_1} = \frac{1}{a_1 b_2 - a_2 b_1}$$

$$\left. \begin{array}{l} ax + by + c_1 = 0 \\ ax + by + c_2 = 0 \end{array} \right\} \text{দূরত্ব } MN = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

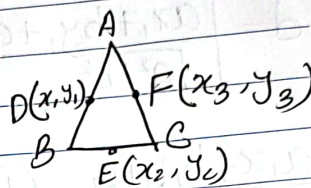
$$P(r_1, \theta_1), Q(r_2, \theta_2)$$

$$PQ = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)}$$

Note

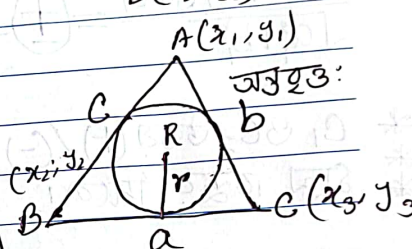


$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} [(x_1 - x_2)(y_2 - y_3) - (y_1 - y_2)(x_3 - x_2)]$$



$$A = (x_1 + x_3 - x_2, y_1 + y_3 - y_2)$$

$$R = \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right)$$

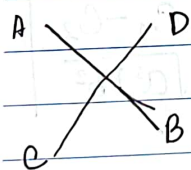


$$S_{ABC} = S_{ABC} = r(a+b+c)$$

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

$$\textcircled{I} ax + by \pm d\sqrt{a^2 + b^2} = 0$$

$$\textcircled{II} a(x - x_1) + b(y - y_1) \pm r\sqrt{a^2 + b^2} = 0$$

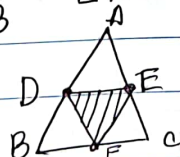


$$\frac{\Delta ABC}{\Delta ABD} = \frac{a}{a}$$

অনুপাত

$$\frac{\Delta ABD}{\Delta ABC} < 0 \text{ অসম্ভব}$$

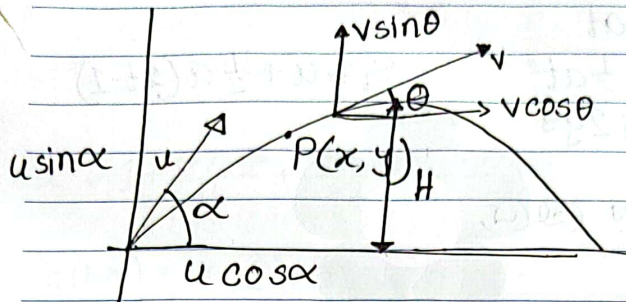
$$> 0 \text{ বহিঃ}$$



$$\Delta ABC = 4 \Delta DEF$$

Note HM-2

Chapter-109



$$x = u t \cos \alpha \quad y = u t \sin \alpha - \frac{1}{2} g t^2$$

$$R = \frac{2u^2 \sin \alpha \cos \alpha}{g}$$

$$T = \frac{2u \sin \alpha}{g}$$

$$H = \frac{u^2 \sin^2 \alpha}{2g}$$

$$t_{\max} = \frac{2u \sin \alpha}{g}$$

$$\tan \alpha = \frac{4H}{R}, \quad T^2 = \frac{8H}{g} = \frac{2R \tan \alpha}{g}$$

$$y = x \tan \alpha \left(1 - \frac{x}{R}\right) = x \tan \left(1 - \frac{gx}{u^2 \sin 2\alpha}\right)$$

Note

$$v = u + at$$

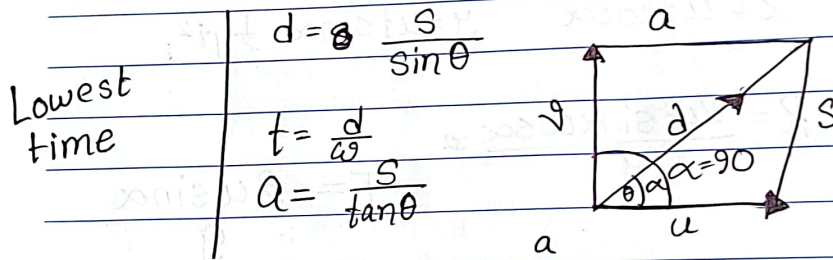
$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s_t = u + \frac{1}{2}a(2t-1)$$

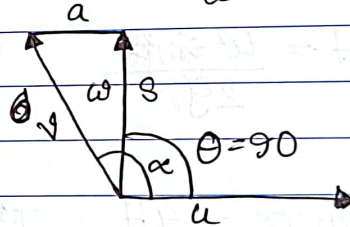
নদী লোম্বা হলে,

$$\alpha = 90^\circ \text{ হলে } w = \sqrt{u^2 + v^2} \quad \theta = \tan^{-1}\left(\frac{v}{u}\right)$$



~~কিন্তু~~ $\theta = 90^\circ$ হলে,

$$\cos \alpha = \frac{u}{w} \Rightarrow u = w \cos \alpha$$



$$w = \sqrt{v^2 + u^2}$$

$$t = \frac{s}{u \sin \theta}$$

Note

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$a^x = 1 + \frac{(x \ln a)^1}{1!} + \frac{(x \ln a)^2}{2!} + \frac{(x \ln a)^3}{3!} + \dots$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\log(1+x)$$

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$$

1. Complete the relationship that is (b) _____ on the teacher-student relationship that is (b) _____ their mother (c) _____ help from them (d) _____ understand form _____ in conformity with the circumstances of the situation.

2. In an educational institution, the teacher-student relationship is (a) _____ (b) _____ (c) _____ (d) _____ (e) _____ (f) _____ (g) _____ (h) _____ (i) _____ (j) _____ (k) _____ (l) _____ (m) _____ (n) _____ (o) _____ (p) _____ (q) _____ (r) _____ (s) _____ (t) _____ (u) _____ (v) _____ (w) _____ (x) _____ (y) _____ (z) _____

Note M-1

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} a^x = a^x \ln a$$

$$\frac{d}{dx} c = 0$$

$$\frac{d}{dx} e^{mx} = m e^{mx}$$

$$y = x^z$$

$$\frac{d}{dx} \log_a x = \frac{1}{x} \log_a e$$

$$y_1 = y \left[z \cdot \frac{1}{x} + z_1 \ln x \right]$$

Note

$$\int \sin x \, dx = -\cos x$$

$$\int \cos x \, dx = \sin x$$

$$\int \sec x \, dx = \ln |\sec x + \tan x|$$

$$\int \operatorname{cosec} x \, dx = \ln |\operatorname{cosec} x - \cot x|$$

$$\int \tan x \, dx = \ln |\sec x|$$

$$\int \cot x \, dx = \ln |\sin x|$$

$$\int \sec^2 x \, dx = \tan x$$

$$\int \sec x \tan x \, dx = \sec x$$

$$\int u v \, dx = u \int v \, dx - \int \left[\frac{d}{dx} u (\int v \, dx) \right] dx$$

$$\int \ln x \, dx = x \ln x - x + c$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}$$

Note

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$$

$$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \frac{1}{a} \sin^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln \left| x + \sqrt{x^2-a^2} \right|$$

$$\int \frac{1}{\sqrt{a^2+x^2}} dx = \ln \left| x + \sqrt{x^2+a^2} \right|$$

$$\int \sqrt{a^2-x^2} dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

Note

$$\int \sqrt{a^2 + x^2} dx = \frac{x\sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \ln |x + \sqrt{a^2 + x^2}|$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}|$$

$$\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + c$$

$$y = e^{mx+c} \parallel y_n = m^n e^{mx+c}$$

$$y = x^m \parallel y_n = {}^m P_n x^{m-n} [m \geq n]$$
$$y_n = 0 [m < n]$$

$$y = a \sin(bx+c) \parallel y_n = a b^n \sin(n\pi/2 + bx+c)$$

$$y = a \cos(bx+c) \parallel y_n = a b^n \cos(n\pi/2 + bx+c)$$

$$y = e^{ax} \sin(bx+c) \parallel y_n = e^{ax} r^n \sin(n\theta + bx+c)$$

$$y = e^{ax} \cos(bx+c) \parallel y_n = e^{ax} r^n \cos(n\theta + bx+c)$$

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}(b/a)$$

$$y = \frac{1}{ax+b} \parallel y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

Note

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{1}{a} \sin^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \ln \left| x + \sqrt{x^2-a^2} \right|$$

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \ln \left| x + \sqrt{a^2+x^2} \right|$$

Note

$$\int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$\int \sqrt{a^2 + x^2} dx = \frac{x\sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \ln |x + \sqrt{x^2 + a^2}|$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}|$$

$$y = \ln(ax+b) \quad y_n = \frac{n! a^n (-1)^n}{(ax+b)^n}$$

1. Complete the following sentences:
In an educational institution, the relationship that is (a) _____ but (c) _____ helps from them in (b) _____ on the basis of their means (d) _____ understand / form consider

Note

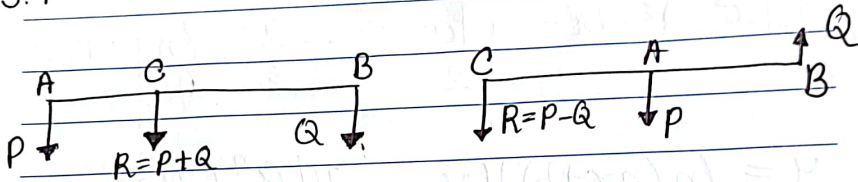
Chpt-08

HM-2

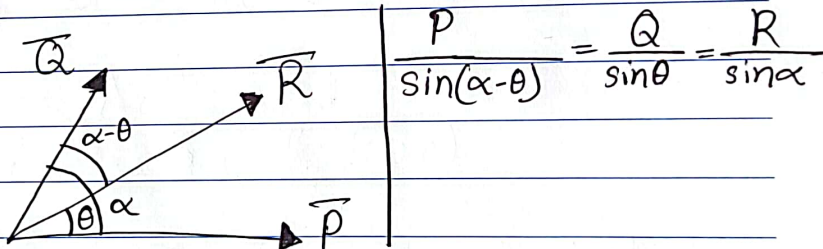
1. $P=Q$ হলে $R=2P\cos\frac{\alpha}{2}$ $\theta=\frac{\alpha}{2}$

2. $R \perp P$ then $R^2=Q^2-P^2$ $\cos\alpha=-\frac{Q}{P}$

3. $P \times m$ করলে $R \times m$ হলে, $\cos\alpha=-\frac{(m+1)Q}{2mP}$



$$\frac{P}{BC} = \frac{Q}{AC} = \frac{R}{AB}$$



কোনো দুটি বা ততোধিক বল একত্রিত হলে একটি বিন্দুতে

দাঁড়ি = $\sqrt{3}$ $d \rightarrow$ অন্তর

Note HM-1

Chpt-9

$$1. \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{x^n \text{ ଏବଂ ସହଜ}}{x^n \text{ ଏବଂ ସହଜ}}$$

Ex:

$$\lim_{x \rightarrow \infty} \frac{0x^3 + 5x^2 + 6x + 5}{7x^3 - 2x + 9} = \frac{0}{7} = 0$$

$$2. \lim_{x \rightarrow 0} (1+ax)^{b/x} = \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^{bx}$$

$$= e^{x \text{ ଏବଂ ସହଜ} \times \frac{1}{x} \text{ ଏବଂ ସହଜ}}$$

Ex

$$\lim_{x \rightarrow 0} (1+7x)^{\frac{5x+3}{x}} = \lim_{x \rightarrow 0} (1+7x)^{(5 + 3/x)}$$

$$= \lim_{x \rightarrow 0} (1+7x)^5 \cdot \lim_{x \rightarrow 0} (1+7x)^{3/x}$$

$$= e^{7 \cdot 3} = e^{21}$$

1. Complete the following sentences by writing the words in the box in the correct form.
In an educational institution teacher-student relationship that is (a) _____ but (c) _____ help from them on the basis of their memory in case they (h) _____
2. In conformity with the circumstances of the situation, the teacher should (i) _____ understand / form / consider / take
3. In conformity with the circumstances of the situation, the teacher should (i) _____ understand / form / consider / take

Note

$$3. \lim_{x \rightarrow 0} \frac{\sin Ax - \sin Bx}{\sin Cx - \sin Dx} = \frac{A-B}{C-D}$$

$$4. \lim_{x \rightarrow 0} \frac{\cos Ax - \cos Bx}{\cos Cx - \cos Dx} = \frac{B^2 - A^2}{D^2 - C^2}$$

5. $y = x^n$	$y_{n-1} = \cancel{(n-1)x^{n-1}} n! x$
$y_n = \cancel{n!}$	$\emptyset$
$y_{n+1} = 0$	

$$6. \begin{aligned} y &= \sin ax \\ y_n &= a^n \sin(n\pi/2 + ax) \end{aligned}$$

$$7. \begin{aligned} y &= \cos bx \\ y_n &= b^n \cos(n\pi/2 + bx) \end{aligned}$$

$$\left\{ \begin{aligned} &\text{Area of Eclipse} = \pi ab \\ &\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \end{aligned} \right.$$

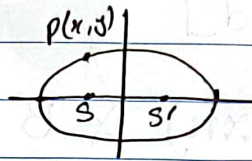
$$8. \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

Note HM-2 Chpt-6

i) $y^2 = 4ax$ (x, y) ଚିହ୍ନିତେ ଯକ୍ଷର, $yy_1 = 2a(x+x_1)$

ii) $x^2 = 4ay$ (x, y) ଚିହ୍ନିତେ ଯକ୍ଷର, $xx_1 = 2a(y+y_1)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



$y = mx + c$ ଯକ୍ଷର $c = \pm \sqrt{a^2 m^2 + b^2}$

ଯକ୍ଷଚିହ୍ନ $\left(\frac{-a^2 m}{c}, \frac{b^2}{c} \right)$

ସ୍ପର୍ଶକ $SP + S'P = 2a$ [$a > b$]

x, y ଚିହ୍ନିତେ ଯକ୍ଷର,

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

~~ଫୋକାଲ ଚାପ~~

ଜ୍ୟା (x, y) ଚିହ୍ନିତେ ସମୀକ୍ଷିତ ହେଉ, ଜ୍ୟା ସମୀ:

$$b^2 xx_1 + a^2 yy_1 = b^2 x_1 + a^2 y_1$$

1. Complete the following sentences using the words in the box.
2. In an educational institution that is (a) help from them in (b) on the basis of (c) their memory (d) understand (e) form (f) consider (g) relationship that is (h) but (i) they (j) they (k) they (l) they (m) they (n) they (o) they (p) they (q) they (r) they (s) they (t) they (u) they (v) they (w) they (x) they (y) they (z) they (aa) they (ab) they (ac) they (ad) they (ae) they (af) they (ag) they (ah) they (ai) they (aj) they (ak) they (al) they (am) they (an) they (ao) they (ap) they (aq) they (ar) they (as) they (at) they (au) they (av) they (aw) they (ax) they (ay) they (az) they (ba) they (bb) they (bc) they (bd) they (be) they (bf) they (bg) they (bh) they (bi) they (bj) they (bk) they (bl) they (bm) they (bn) they (bo) they (bp) they (bq) they (br) they (bs) they (bt) they (bu) they (bv) they (bw) they (bx) they (by) they (bz) they (ca) they (cb) they (cc) they (cd) they (ce) they (cf) they (cg) they (ch) they (ci) they (cj) they (ck) they (cl) they (cm) they (cn) they (co) they 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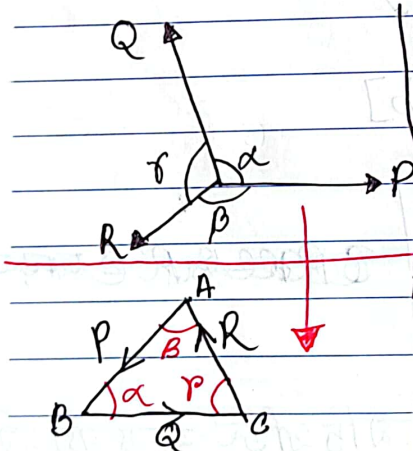
Note HM-1 Chpt-1

$$1. \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$2. \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ সন্মতি (} A^2 = A \text{) যদি } a+d=1 \text{ এবং } bc = a-a^2 = d-d^2$$

$$3. A (m \times n) \times B (n \times 1) \text{ হলে } AB = m \times 1$$

HM-2 Chpt-8 Ext



$$\frac{P}{\sin r} = \frac{Q}{\sin \beta} = \frac{R}{\sin \alpha}$$

$$\cos \alpha = \frac{R^2 - P^2 - Q^2}{2PQ}$$

$$\cos \beta = \frac{Q^2 - P^2 - R^2}{2PR}$$

$$\cos \gamma = \frac{P^2 - Q^2 - R^2}{2QR}$$

$$\frac{P}{AB} = \frac{Q}{BC} = \frac{R}{CA}$$

Note

$\vec{P} \vec{Q} \vec{R}$

০ বিকল্পে OA, OB, OC সমান্তরাল থাকবে,

০ অঙ্ক: ত্রু (A)

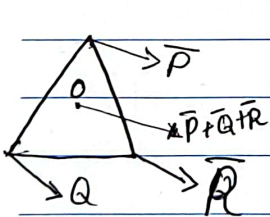
$$P:Q:R = \cos \frac{A}{2} : \cos \frac{B}{2} : \cos \frac{C}{2}$$

$$P^2:Q^2:R^2 = a(b+c-a) : b(c+a-b) : c(a+b-c)$$

০ মধ্যস্থিত (A) $P:Q:R = \sin A : \sin B : \sin C$
 $= a : b : c$

০ পরিবর্তন (A) $P:Q:R = \sin 2A : \sin 2B : \sin 2C$
 $= a \cos A : b \cos B : c \cos C$

$$P:Q:R = a^2(b^2+c^2-a^2) : b^2(c^2+a^2-b^2) : c^2(a^2+b^2-c^2)$$



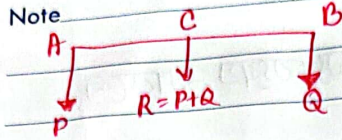
০ ত্রুত্ব $P=Q=R$

০ মধ্যস্থিত $P:Q:R = \tan A : \tan B : \tan C$

০ অঙ্কত্ব $= \sin A : \sin B : \sin C$

০ পরিবর্তন $= \sin 2A : \sin 2B : \sin 2C$

Note



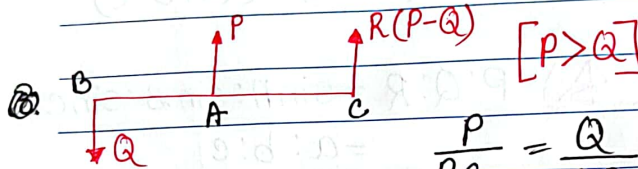
$$P \cdot AC = Q \cdot BC$$

$$AB = d$$

$$\frac{P}{BC} = \frac{Q}{AC} = \frac{R}{AB}$$

1. P ଓ Q ଦୁହେଁ ସମାନେ ବାହାରିବା $\frac{P \times Q}{P+Q}$ ଦୂରରେ ରହେ

2. P ଓ Q ଅସମାନ ଚିହ୍ନିତ କଲେ ବାହାରିବା $\frac{P-Q}{P+Q} \cdot d$ ଦୂରରେ ରହେ



$$\frac{P}{BC} = \frac{Q}{AC} = \frac{R}{AB}$$

P ଓ Q ଧନ ଚିହ୍ନିତ କଲେ ବାହାରିବା $\frac{P-Q}{P+Q} \cdot AB$

ଦୂରରେ ରହେ।